Longest Palindromic Substring

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The problem

- Expand around every centre (odd and even).

Intuition

- Every palindrome has a centre; expand around each of the 2n minus 1 centres.

Approach

- 1. For each index, expand odd and even centres
- 2. Track the longest span
- 3. Return the substring

Java solution

```java
public class LongestPalindromicSubstring {
    public static String longestPalindromicSubstring(String s) {
        int bs = 0, be = 0;
        for (int i = 0; i < s.length(); i++)
            for (int j = i; j <= i + 1; j++) {
                int l = i, r = j;
                while (l >= 0 && r < s.length() && s.charAt(l) == s.charAt(r)) { l--; r++; }
                if (r - l - 1 > be - bs) { bs = l + 1; be = r; }
            }
        return s.substring(bs, be);
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n) output

Recap

- Every palindrome has a centre; expand around each of the 2n minus 1 centres.
- Complexity: O(n^2)